

cross_classified_models

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```
library(arrow)
library(lme4)
library(lmerTest)
library(dplyr)
library(performance)
library(broom.mixed)
library(ggplot2)
library(jtools) # new package
library(tidyverse)

options(scipen = 999)
```

```
# Setting repository path
#repo_path <- "/Users/namomac/Desktop/Moneyball_FC/inputs/"
repo_path <- "D:/repos/Moneyball_FC/inputs/"

# Reading data
dat <- read_parquet(str_c(repo_path, "player_df.parquet"))
#dat <- read_parquet("player_df.parquet")

# View the data
glimpse(dat)
```

Rows: 7,168

Columns: 27

\$ season_year	<dbl> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013~
\$ club_id	<dbl> 11, 11, 11, 11, 11, 11, 11, 11, 11, 11, 11, 11~
\$ player_id	<dbl> 132, 5293, 6710, 7451, 15185, 15378, 15799, 15~
\$ yellow_cards	<dbl> 3, 1, 4, 8, 3, 1, 4, 5, 1, 2, 3, 0, 2, 3, 0, 2~
\$ red_cards	<dbl> 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~

```

$ goals <dbl> 3, 0, 3, 6, 16, 0, 12, 0, 0, 0, 0, 0, 7, 16, 0~
$ assists <dbl> 1, 0, 0, 6, 12, 1, 13, 0, 0, 1, 0, 0, 5, 13, 0~
$ minutes_played <dbl> 904, 90, 3838, 3795, 2610, 132, 4131, 3047, 90~
$ name <chr> "Tomas Rosicky", "Sébastien Squillaci", "Per M~
$ country_of_birth <chr> "CSSR", "France", "Germany", "Spain", "Poland"~
$ country_of_citizenship <chr> "Czech Republic", "France", "Germany", "Spain"~
$ date_of_birth <dtm> 1980-10-04, 1980-08-11, 1984-09-29, 1982-03-2~
$ sub_position <chr> "Attacking Midfield", "Centre-Back", "Centre-B~
$ position <chr> "Midfield", "Defender", "Defender", "Midfield"~
$ height_in_cm <dbl> 179, 185, 200, 177, 182, 172, 168, 183, 188, 1~
$ market_value_ave <dbl> 2000000.0, 833333.3, 16000000.0, 10000000.0, 2~
$ Age <dbl> 33, 33, 29, 31, 28, 32, 29, 28, 29, 27, 30, 28~
$ Club <chr> "Arsenal", "Arsenal", "Arsenal", "Arsenal", "A~
$ Country <chr> "England", "England", "England", "England", "E~
$ year_ex_rate <dbl> 1.33, 1.33, 1.33, 1.33, 1.33, 1.33, 1.33, 1.33~
$ market_value_ave_usd <dbl> 2660000, 1108333, 21280000, 13300000, 30590000~
$ usd_revenue_2024 <dbl> 3581828.4, 1492428.5, 28654627.2, 17909142.0, ~
$ foreign_players <dbl> 0.8214286, 0.8214286, 0.8214286, 0.8214286, 0.~
$ uefa_coefficient <dbl> 15, 15, 15, 15, 15, 15, 15, 15, 15, 15, 15, 15~
$ country_of_birth_fixed <chr> "Czechoslovak Socialist Republic", "France", "~
$ Region <chr> "Europe & Central Asia", "Europe & Central Asi~
$ citizen_of_club_country <chr> "No", "No", "No", "No", "No", "No", "No", "No"~

```

Data Structure

Level 1 (time): season_year: season-year within player Level 2 (player): name: players Level 3 (club): club_id: clubs

L1 time-varying: season_year, yellow_cards, red_cards, goals, assists, minutes_played L2 player-invariant: country_of_birth, sub_position, position, height_in_cm L3 club-invariant: season_year

season_year: 2013, 2014, 2015, 2016, 2017, 2018, 2020, 2021, 2022, 2023, 2024.

position: Attack, Defender, Goalkeeper, Midfield.

Region: East Asia & Pacific, Europe & Central Asia, Latin America & Caribbean, Middle East & North Africa, North America, Sub-Saharan Africa.

citizen_of_club_country: YES, NO.

```

player_df <- dat %>%
  mutate(
    player_id = factor(player_id),
    club_id   = factor(club_id),
    position = factor(position),
    Region = factor(Region),
    citizen_of_club_country = factor(citizen_of_club_country),
    usd_revenue_2024_log = log(usd_revenue_2024),
    # convert season_year to 0, 1, 2, etc.
    season_year_c = season_year - min(season_year)
  )

```

- Check missing values.

```

# Which variable has missing value
colSums(is.na(player_df))

```

season_year	club_id	player_id
0	0	0
yellow_cards	red_cards	goals
0	0	0
assists	minutes_played	name
0	0	0
country_of_birth	country_of_citizenship	date_of_birth
108	103	2
sub_position	position	height_in_cm
0	0	36
market_value_ave	Age	Club
0	2	0
Country	year_ex_rate	market_value_ave_usd
0	0	0
usd_revenue_2024	foreign_players	uefa_coefficient
0	0	0
country_of_birth_fixed	Region	citizen_of_club_country
108	108	103
usd_revenue_2024_log	season_year_c	
0	0	

```

# Which observation has missing values
player_df %>%
  filter(!complete.cases())

```

```
# A tibble: 145 x 29
  season_year club_id player_id yellow_cards red_cards goals assists
      <dbl> <fct>   <fct>         <dbl>      <dbl> <dbl>   <dbl>
1      2013 12     4311             1          0     0       0
2      2013 15     1073             3          0     2       0
3      2013 15    43274            12          0     3       1
4      2013 15    51168             3          0     5       0
5      2013 15    53173             7          0     3       0
6      2013 16    58124             5          0     5       2
7      2013 27     2514            10          0     9      11
8      2013 46    28021             4          0     0       0
9      2013 148    3476             1          0     0       0
10     2013 148    4192             0          0     0       0
# i 135 more rows
# i 22 more variables: minutes_played <dbl>, name <chr>,
#   country_of_birth <chr>, country_of_citizenship <chr>, date_of_birth <dtm>,
#   sub_position <chr>, position <fct>, height_in_cm <dbl>,
#   market_value_ave <dbl>, Age <dbl>, Club <chr>, Country <chr>,
#   year_ex_rate <dbl>, market_value_ave_usd <dbl>, usd_revenue_2024 <dbl>,
#   foreign_players <dbl>, uefa_coefficient <dbl>, ...

# Exclude all missing values, only 7023 observations now
clean_player_df <- na.omit(player_df)
```

- Intercept only model: We want to model usd_revenue_2024 because market value is accounting for inflation and converted to 2024 US dollars.

Fit the model

Cross-classified

- The player went to different clubs across years, thus they are cross-clubs, not nested in one club.
- So we use (1 | club_id) + (1|player_id) here, not (1 | club_id) + (1|club_id : player_id) here, since the latter one is used only when player are nested in the clubs.

```
clean_player_df %>%
  distinct(player_id, club_id) %>%
  count(player_id, name = "num_clubs") %>%
  filter(num_clubs > 1) %>%
  arrange(desc(num_clubs))
```

```
# A tibble: 441 x 2
  player_id num_clubs
  <fct>      <int>
1 182712      7
2 45320      5
3 55735      5
4 119296      5
5 282429      5
6 27389      4
7 35564      4
8 37389      4
9 37666      4
10 39153      4
# i 431 more rows
```

We will now add a fixed effect for year and allow it to vary by club.

- Recall that year has been centered at 0, so that 2013=0, 2024 = 11. This is because `lmer` struggles when predictors are on vastly different scales than the outcome. If `season_year` is recorded as “2020, 2021, 2022,” these numbers are huge compared to the log-revenue variance. This causes the gradient calculation to fail.

```
fixed_year <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c +
    # random effects
    (1|player_id) + (1|club_id), ,
  data = clean_player_df, REML = TRUE)

summ(fixed_year)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20052.80
BIC	20087.09
Pseudo-R ² (fixed effects)	0.00
Pseudo-R ² (total)	0.79

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.99	0.07	221.31	32.91	0.00
season_year_c	-0.03	0.00	-8.11	6524.03	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.34
club_id	(Intercept)	0.28
Residual		0.71

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.76
club_id	21	0.03

- We can see that player_id accounts for about 76% of market value while club_id accounts for 3%.
- we also fit a random slope for year to compare.
 - To estimate a random slope (`1 + year_centered | player_id`), the model needs multiple data points for *every* player to calculate their individual trajectory. But many players only appear in our data for 1 or 2 years, and the model struggles to distinguish the “player slope” from the “residual error,” causing these convergence issues. Clubs, however, have many players and many years of historical market value data so a random slope for clubs is more robust to fit.

```
random_year <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c +
    # random effects
    (1|player_id) + (1+season_year_c|club_id),
  data = clean_player_df, REML = TRUE)

summ(random_year)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	19993.96
BIC	20041.96
Pseudo-R ² (fixed effects)	0.00
Pseudo-R ² (total)	0.80

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.98	0.09	187.65	24.67	0.00
season_year_c	-0.03	0.01	-3.17	20.11	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.35
club_id	(Intercept)	0.35
club_id	season_year_c	0.04
Residual		0.70

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.75
club_id	21	0.05

- AIC and BIC values are lower with the random slope of year.

```
calculate_lrt_mixture <- function(reduced_model, full_model) {

  # 1. Extract Log-Likelihoods
  ll_reduced <- as.numeric(logLik(reduced_model))
  ll_full    <- as.numeric(logLik(full_model))

  # 2. Calculate LRT Statistic
  # Formula: -2 * (LL_reduced - LL_full)
```

```

lrt_stat <- -2 * (ll_reduced - ll_full)

# 3. Calculate Difference in Parameters (df)
# This tells us how many params we added
df_reduced <- attr(logLik(reduced_model), "df")
df_full    <- attr(logLik(full_model), "df")
df_diff    <- df_full - df_reduced

# 4. Calculate P-Value based on the difference
# If we added a slope + covariance, df_diff should be 2.
# The mixture is 0.5 * ChiSq(1) + 0.5 * ChiSq(2)

if (df_diff == 2) {
  p_val <- 0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE) +
    0.5 * pchisq(lrt_stat, df = 2, lower.tail = FALSE)

  msg <- "Mixture of ChiSq(1) and ChiSq(2)"
} else if (df_diff == 1) {
  # If we added a parameter with NO covariance (e.g. diag structure)
  p_val <- 0.5 * pchisq(lrt_stat, df = 0, lower.tail = FALSE) +
    0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE)

  msg <- "Mixture of ChiSq(0) and ChiSq(1)"
} else {
  # Fallback for non-standard comparisons
  p_val <- pchisq(lrt_stat, df = df_diff, lower.tail = FALSE)
  msg <- paste0("Standard ChiSq(", df_diff, ")")
}

# Output
cat("LRT Statistic:", round(lrt_stat, 4), "\n")
cat("Parameter Diff:", df_diff, "\n")
cat("Distribution: ", msg, "\n")
cat("P-Value:      ", format.pval(p_val, eps = .0001), "\n")
}

# Run the comparison
calculate_lrt_mixture(fixed_year, random_year)

```

LRT Statistic: 62.841

Parameter Diff: 2
 Distribution: Mixture of ChiSq(1) and ChiSq(2)
 P-Value: < 0.0001

- We use a helper function to print the p-value for the LRT difference, and find a p-value <.0001 and we reject the Null, (with $.5 \times 2(1) + .5 \times 2(2)$ mixture) and conclude that there is strong evidence for club variation in market value trajectories and therefore we should keep this random slope in the model.

```
compare_performance(fixed_year, random_year, metrics = c("AIC", "BIC"))
```

Comparison of Model Performance Indices

Name	Model	AIC (weights)	BIC (weights)
fixed_year	lmerModLmerTest	20040.0 (<.001)	20074.3 (<.001)
random_year	lmerModLmerTest	19982.9 (>.999)	20030.9 (>.999)

Adding first level player and club predictors

- We first want to group center minutes played for each for each club by year. We also need to account for the quadratic relationship between age and market value, by centering age by year by club.

```
clean_player_df <- clean_player_df |>
  group_by(season_year_c, club_id) |>
  mutate(minutes_played_c = minutes_played - mean(minutes_played, na.rm=TRUE),
         age_c = (Age - mean(Age, na.rm=TRUE))^2) |>
  ungroup()
```

```
player_varying_predictors <- lmer(
  usd_revenue_2024_log ~ season_year_c +
    # first level predictors
    yellow_cards + red_cards + goals +
    assists + minutes_played + age_c +
    # random effects
    (1|player_id) + (season_year_c|club_id),
```

```
data = clean_player_df, REML = TRUE)

summ(player_varying_predictors)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	15641.28
BIC	15730.42
Pseudo-R ² (fixed effects)	0.28
Pseudo-R ² (total)	0.86

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.82	0.07	223.44	25.40	0.00
season_year_c	0.01	0.01	1.75	20.45	0.10
yellow_cards	0.00	0.00	0.61	5306.45	0.54
red_cards	-0.01	0.02	-0.48	4695.40	0.63
goals	0.01	0.00	3.75	5527.58	0.00
assists	0.01	0.00	4.10	5062.36	0.00
minutes_played	0.00	0.00	24.83	5593.88	0.00
age_c	-0.02	0.00	-51.05	6232.32	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.00
club_id	(Intercept)	0.29
club_id	season_year_c	0.03
Residual		0.51

We next add first level club predictors

- We want to center player height in inches across years. We also want to center uefa coefficients by year.

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.74
club_id	21	0.06

```
clean_player_df <- clean_player_df |>
  group_by(season_year_c) |>
  mutate(height_in_cm_c = height_in_cm - mean(height_in_cm, na.rm=TRUE),
         uefa_coefficient_c = uefa_coefficient -
           mean(uefa_coefficient)) |>
  ungroup()
```

```
player_club_varying_predictors <- lmer(
  usd_revenue_2024_log ~ season_year_c +
    # first level player predictors
    yellow_cards + red_cards + goals +
    assists + minutes_played + age_c +
    # first level club predictors
    height_in_cm_c + foreign_players + uefa_coefficient_c +
    # second level predictors
    position + Region + citizen_of_club_country+
    # random effects
    (1|player_id) + (1|club_id),
  data = clean_player_df, REML = TRUE)

summ(player_club_varying_predictors)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	15533.63
BIC	15691.34
Pseudo-R ² (fixed effects)	0.36
Pseudo-R ² (total)	0.86

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.55	0.22	70.89	1590.63	0.00
season_year_c	0.01	0.00	3.60	6703.56	0.00
yellow_cards	-0.00	0.00	-0.07	5229.11	0.94
red_cards	-0.01	0.02	-0.55	4702.43	0.58
goals	0.01	0.00	2.94	5322.45	0.00
assists	0.01	0.00	3.69	4972.49	0.00
minutes_played	0.00	0.00	26.39	5465.39	0.00
age_c	-0.02	0.00	-50.10	6300.98	0.00
height_in_cm_c	0.01	0.00	3.41	1766.08	0.00
foreign_players	0.38	0.10	3.96	5125.48	0.00
uefa_coefficient_c	-0.00	0.00	-0.71	4975.27	0.48
positionDefender	-0.33	0.06	-5.65	1862.91	0.00
positionGoalkeeper	-0.90	0.09	-9.72	1814.23	0.00
positionMidfield	-0.15	0.06	-2.54	1808.07	0.01
RegionEurope & Central Asia	0.29	0.20	1.46	1795.03	0.14
RegionLatin America & Caribbean	0.55	0.21	2.67	1785.15	0.01
RegionMiddle East & North Africa	0.24	0.31	0.79	1766.50	0.43
RegionNorth America	0.21	0.29	0.73	1800.08	0.47
RegionSub-Saharan Africa	0.36	0.22	1.64	1783.30	0.10
citizen_of_club_countryYes	-0.25	0.03	-7.48	6500.43	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.93
club_id	(Intercept)	0.25
Residual		0.52

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.72
club_id	21	0.05

Adding level 2 predictors

- We want to center height in inches across years and clubs.

Why This Syntax Works for Cross-Classification

In `lme4`, you do not need special “cross-classified” syntax commands. The structure is defined by how you request the random effects:

1. **Cross-Classification:** By using `(1 | player_id) + (1 | club_id)`, you are telling R that `player_id` is **not** a subset of `club_id`. R treats them as two independent sources of variation that cross over each other (since players move between clubs).
2. **The Random Slope:** By using `(1 + season_year_c | club_id)`, you are specifically testing the hypothesis that **Club A** might be great at increasing player value over time (steep positive slope), while **Club B** might be stagnant (flat slope), even if they start with different intercepts.

```
#Using bobyqa to handle the complex random effects structure and avoid gradient warnings
strict_control <- lmerControl(optimizer = "bobyqa",
                              optCtrl = list(maxfun = 200000))

final_fit <- lmer(
  usd_revenue_2024_log ~
    # --- Fixed Effects: Time ---
    season_year_c +

    # --- Fixed Effects: Player Level ---
    yellow_cards +
    red_cards +
    goals +
    assists +
    minutes_played_c + # centered by year & club
    age_c +           # centered by year & club

    # --- Fixed Effects: Club Level ---
    height_in_cm_c +   # centered by year
    foreign_players +
    uefa_coefficient_c + # centered by year

    # --- Fixed Effects: Categorical ---
    position +
    Region +
```

```

citizen_of_club_country +

# --- Random Effects (Cross-Classified) ---
(1 | player_id) +
(1 + season_year_c | club_id),
data = clean_player_df, REML = TRUE, control = strict_control)

summ(final_fit)

```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	15509.74
BIC	15681.17
Pseudo-R ² (fixed effects)	0.35
Pseudo-R ² (total)	0.86

1. The Mathematical Equation (LaTeX)

This is a **cross-classified multilevel model** with random intercepts for players and random slopes for clubs.

Let: * t index the **Time** (Season Year). * i index the **Player**. * j index the **Club**.

The log-transformed market value Y_{tij} is modeled as:

$$\begin{aligned}
Y_{tij} = & \beta_0 + \beta_1(\text{Year}_{tij}) \\
& + \underbrace{\beta_2 \text{Goals}_{tij} + \beta_3 \text{Assists}_{tij} + \dots + \beta_k(\text{Age}_{tij} - \bar{\text{Age}}_{tj})}_{\text{Player Fixed Effects (Time-Varying)}} \\
& + \underbrace{\beta_m(\text{Height}_{tij} - \bar{\text{Height}}_{tj}) + \beta_n \text{Foreign}_{tj} + \beta_p(\text{UEFA}_{tj} - \bar{\text{UEFA}}_t)}_{\text{Club/Context Fixed Effects}} \\
& + \underbrace{\beta_q \text{Pos}_i + \beta_r \text{Region}_i + \dots}_{\text{Time-Invariant Controls}} \\
& + \underbrace{u_{0i}}_{\text{Player RE}} + \underbrace{v_{0j} + v_{1j}(\text{Year}_{tij})}_{\text{Club RE}} + \epsilon_{tij}
\end{aligned}$$

Where the Random Effects are distributed as:

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.93	0.23	69.26	941.99	0.00
season_year_c	0.01	0.01	0.77	20.39	0.45
yellow_cards	0.00	0.00	0.70	5187.98	0.49
red_cards	-0.01	0.02	-0.55	4674.31	0.58
goals	0.01	0.00	3.16	5279.82	0.00
assists	0.01	0.00	4.47	4930.09	0.00
minutes_played_c	0.00	0.00	25.97	5391.80	0.00
age_c	-0.02	0.00	-50.27	6274.92	0.00
height_in_cm_c	0.01	0.00	3.50	1769.36	0.00
foreign_players	0.61	0.11	5.73	3350.38	0.00
uefa_coefficient_c	-0.00	0.00	-1.16	4779.52	0.25
positionDefender	-0.32	0.06	-5.52	1852.48	0.00
positionGoalkeeper	-0.89	0.09	-9.53	1805.01	0.00
positionMidfield	-0.14	0.06	-2.43	1802.89	0.02
RegionEurope & Central Asia	0.29	0.20	1.42	1797.26	0.16
RegionLatin America & Caribbean	0.54	0.21	2.60	1788.02	0.01
RegionMiddle East & North Africa	0.23	0.31	0.73	1768.25	0.47
RegionNorth America	0.22	0.29	0.75	1799.53	0.45
RegionSub-Saharan Africa	0.35	0.22	1.55	1786.06	0.12
citizen_of_club_countryYes	-0.24	0.03	-7.28	6542.70	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.94
club_id	(Intercept)	0.35
club_id	season_year_c	0.03
Residual		0.51

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.70
club_id	21	0.10

$$\begin{aligned}
u_{0i} &\sim N(0, \sigma_{\text{player}}^2) && \text{(Player Intercept)} \\
\begin{pmatrix} v_{0j} \\ v_{1j} \end{pmatrix} &\sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_{\text{club_int}}^2 & \sigma_{\text{cov}} \\ \sigma_{\text{cov}} & \sigma_{\text{club_slope}}^2 \end{pmatrix} \right) && \text{(Club Intercept \& Slope)} \\
\epsilon_{tij} &\sim N(0, \sigma_{\epsilon}^2) && \text{(Residual)}
\end{aligned}$$

2. Interpretation of the Final Model

Here is how you interpret the specific coefficients, especially given the custom centering you applied.

A. Fixed Effects (The “Average” Effects)

- **Intercept** (β_0): The expected log market value for a baseline player (likely a Forward/Midfielder from the reference Region) who has **average performance**, is of **average age relative to their team**, and plays for an **average club** at the starting year ($t = 0$).
- **Season Year** (β_1): This is the **global trend**. It tells you the average annual growth rate in player market value across the entire dataset, holding performance constant. A positive value means the market is inflating generally.
- **Player Performance (Goals, Assists, etc.)**: These are standard interpretations. For every 1 additional goal scored, market value increases by β_2 (or $\approx \beta_2 \times 100\%$ since the outcome is logged), holding all else constant.
- $\text{Age}_{tij} - \bar{\text{Age}}_{tj}$ (Group-Mean Centered): **Crucial Distinction**: Because you centered by club and year, this does *not* measure the effect of getting older in general. It measures the effect of being **older relative to your teammates**.
 - *Positive Coeff*: “Being the veteran on the team increases your value.”
 - *Negative Coeff*: “Being older than your average teammate decreases your value” (likely true for soccer players past their prime).
- $\text{UEFA}_{tj} - \bar{\text{UEFA}}_t$ (Year-Centered): This measures the prestige of the club **relative to other clubs that season**. By centering by year, you removed the “grade inflation” of UEFA coefficients over time. A positive coefficient means playing for a club that is currently outperforming the European average boosts a player’s value.

B. Random Effects (The “Variance” Components)

- **Player Random Intercept** (u_{0i}): This captures **innate player quality** (“Superstar Effect”). Even if two players have the exact same goals, assists, and age, Player A might consistently have a higher market value than Player B due to unmeasured factors (marketing potential, leadership, brand).
 - *High σ^2_{player}* : Suggests market value is driven heavily by who the player *is*, not just what they did this year.
- **Club Random Intercept** (v_{0j}): This captures the **baseline club premium**. Playing for Real Madrid or Man City likely adds a baseline value to a player regardless of their performance.
- **Club Random Slope** (v_{1j}): This is the most interesting part. It captures **Club Development Trajectories**.
 - A club with a *high positive random slope* ($v_{1j} > 0$) is an “incubator” or “value creator.” Players at this club see their market value rise faster over time than the global average.
 - A club with a *negative random slope* ($v_{1j} < 0$) might be a place where player values tend to stagnate or decline faster than the market.
- **Covariance** (σ_{cov}): This tells you the relationship between the Club Intercept and Slope.
 - *Positive Correlation*: Rich clubs (high intercept) are also better at growing player value (high slope).
 - *Negative Correlation*: Rich clubs start high but grow slowly (diminishing returns), while smaller clubs start low but grow player value quickly (development clubs).

```
export_summs(random_year, player_varying_predictors,
              player_club_varying_predictors,
              digits=3, r.squared=FALSE, confint = TRUE,
              error_format = "[{conf.low}, {conf.high}]",
              model.names = c("M1", "M2", "M3"))
```

```
export_summs(final_fit,
              digits=3, r.squared=FALSE, confint = TRUE,
              error_format = "[{conf.low}, {conf.high}]",
              model.names = c("M4"))
```

If we want to back log transform we may need to use the broom package.

```
library(broom)

final_mod_back_transform <- augment(final_fit, clean_player_df)

augmented_data <- final_mod_back_transform |>
  mutate(
    fitted_original_scale = exp(.fitted),
  )

augmented_data |>
  select(usd_revenue_2024, fitted_original_scale) |>
  head()
```

```
# A tibble: 6 x 2
  usd_revenue_2024 fitted_original_scale
      <dbl>          <dbl>
1      3581828.      3829181.
2      1492428.      1731085.
3      28654627.     27224914.
4      17909142.     18506097.
5      41191027.     52717810.
6       6715928.      6141327.
```

3. Model Diagnostics

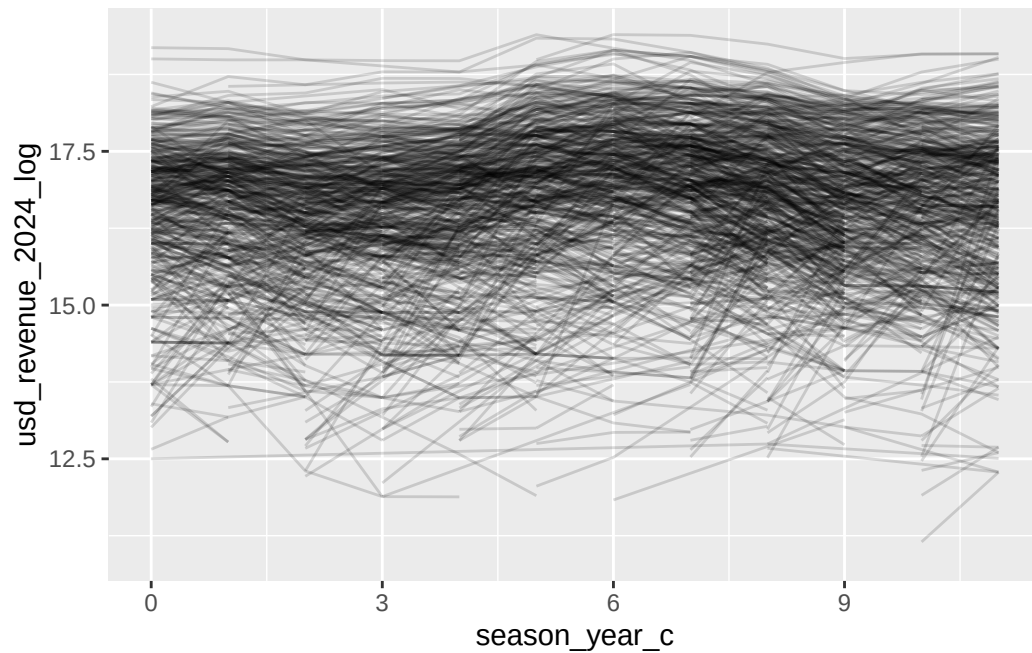
Linearity

In the overall spaghetti plot, the lines vary widely across players, showing considerable between-player heterogeneity in log market value. Within most players, the slopes appear mostly flat, but some of them change up and down dramatically.

The loess smoother plot shows a non-linear averaged trend (curvature) in log-transformed market value (`usd_revenue_2024_log`) over centered season year (`season_year_c`). We can clearly see a peak at about season year 6 to 7, and the shape of the loess line is a wave pattern.

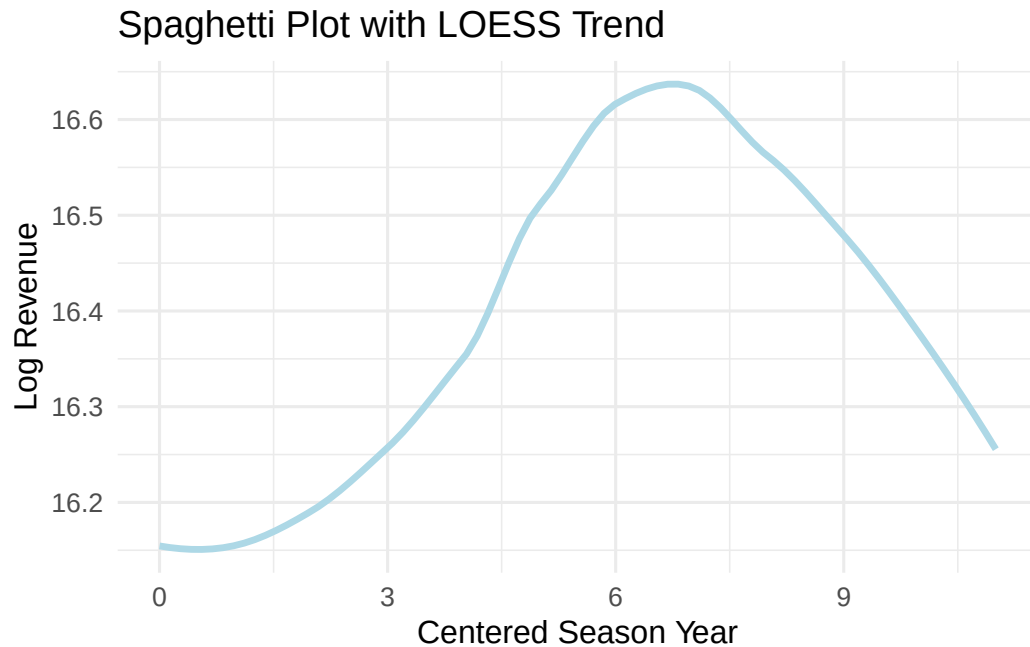
Thus, we can try to fit model with polynomial terms like quadratic and cubic terms of `season_year_c`, instead of assuming linearity.

```
ggplot(clean_player_df) +
  geom_line(aes(x = season_year_c, y = usd_revenue_2024_log, group = player_id), color = "black")
```



```
ggplot(clean_player_df, aes(x = season_year_c,
                           y = usd_revenue_2024_log)) +

  # LOESS smoother (overall trend)
  geom_smooth(method = "loess",
             color = "lightblue", size = 1.2, se = FALSE) +
  labs(x = "Centered Season Year",
       y = "Log Revenue",
       title = "Spaghetti Plot with LOESS Trend") +
  theme_minimal(base_size = 12)
```



Linear, Quadratic, and Curic

By comparing AIC and BIC of these three models, we can conclude that the multilevel model with cubic terms of centered season year can better fit the data. Therefore, we decide to consider cubic term in our final model.

```
final_quad <- lmer(
  usd_revenue_2024_log ~
    # --- Fixed Effects: Time ---
    season_year_c + I(season_year_c^2) +

    # --- Fixed Effects: Player Level ---
    yellow_cards +
    red_cards +
    goals +
    assists +
    minutes_played_c + # centered by year & club
    age_c + # centered by year & club

    # --- Fixed Effects: Club Level ---
    height_in_cm_c + # centered by year
    foreign_players +
    uefa_coefficient_c + # centered by year
```

```

# --- Fixed Effects: Categorical ---
position +
Region +
citizen_of_club_country +

# --- Random Effects (Cross-Classified) ---
(1 | player_id) +
(1 + season_year_c | club_id),
data = clean_player_df, REML = TRUE, control = strict_control)

final_cub <- lmer(
  usd_revenue_2024_log ~
    # --- Fixed Effects: Time ---
    season_year_c + I(season_year_c^2) + I(season_year_c^3) +

    # --- Fixed Effects: Player Level ---
    yellow_cards +
    red_cards +
    goals +
    assists +
    minutes_played_c + # centered by year & club
    age_c + # centered by year & club

    # --- Fixed Effects: Club Level ---
    height_in_cm_c + # centered by year
    foreign_players +
    uefa_coefficient_c + # centered by year

    # --- Fixed Effects: Categorical ---
    position +
    Region +
    citizen_of_club_country +

    # --- Random Effects (Cross-Classified) ---
    (1 | player_id) +
    (1 + season_year_c | club_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

AIC(final_fit, final_quad, final_cub)

```

	df	AIC
final_fit	25	15509.74
final_quad	26	15404.23
final_cub	27	15318.27

```
BIC(final_fit, final_quad, final_cub)
```

	df	BIC
final_fit	25	15681.17
final_quad	26	15582.51
final_cub	27	15503.41

Two level model

In this two level model, we only consider level 1 variables that are time-variant, and level 2 variables that constant within each player.

Here we also center the minutes played, age and height of the players, but only center them by average age and height across all the season years.

We compare AIC and BIC between the two level model with the cubic three level model which consider both player level and club level. The output shows that the two level model better fit the data because both AIC and BIC are lower than the three level model.

```
clean_player_df <- clean_player_df |>
  mutate(minutes_played_c_two = minutes_played - mean(minutes_played, na.rm=TRUE),
         age_c_two = (Age - mean(Age, na.rm=TRUE))^2, height_in_cm_c_two = height_in_cm - me
  ungroup()

final_two <- lmer(
  usd_revenue_2024_log ~
    # --- Fixed Effects: Time ---
    season_year_c + I(season_year_c^2) + I(season_year_c^3) +

    # --- Fixed Effects: Player Level ---
    yellow_cards +
    red_cards +
    goals +
    assists +
    minutes_played_c_two + # centered
    age_c_two + # centered
```

```

# --- Fixed Effects: Club Level ---
height_in_cm_c_two +      # centered by year

# --- Fixed Effects: Categorical ---
position +
Region +
# --- Random Effects (Only Players) ---
(1 | player_id),
data = clean_player_df, REML = TRUE, control = strict_control)

AIC(final_cub, final_two)

```

```

          df      AIC
final_cub 27 15318.27
final_two 21 14925.97

```

```

BIC(final_cub, final_two)

```

```

          df      BIC
final_cub 27 15503.41
final_two 21 15069.96

```

Normality

By comparing the QQ-line and QQ-plot (both residuals and random effects) we can find that both model plots look good on normality, except some outliers. By comparing the residual-fitted value plot, we can find that both model show the same spread pattern, the dots are evenly spread across zero-mean line up and down.

By comparing EBLUPs plots, we can see that most of the points lie deviate from the diagonal at both tails, and the lower side is more downward, suggesting moderate departures from normality in the random slopes and random intercepts, which indicates possible skewness and outliers. Thus, we cannot say the normal random effects assumption in the multilevel model are met.

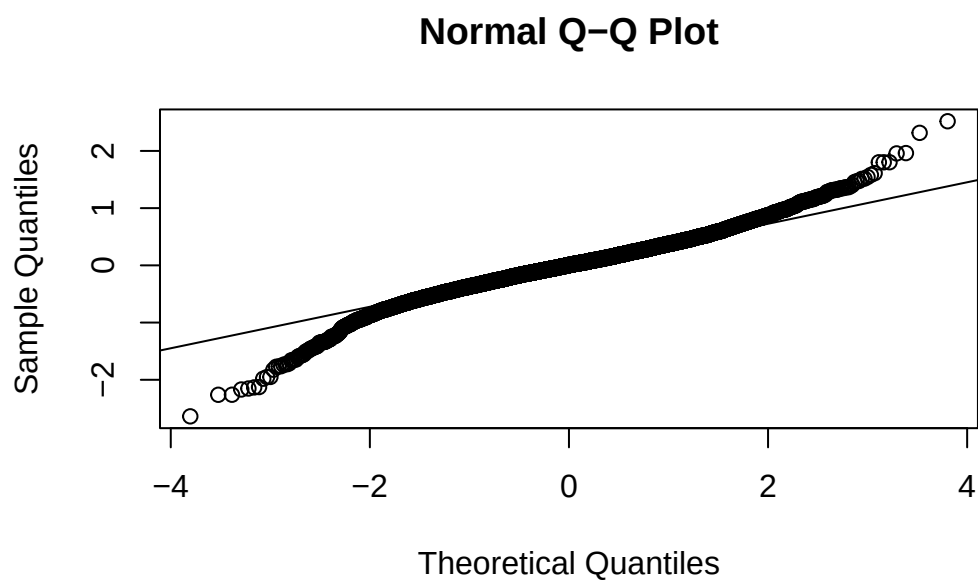
?How to explain the outliers?

```

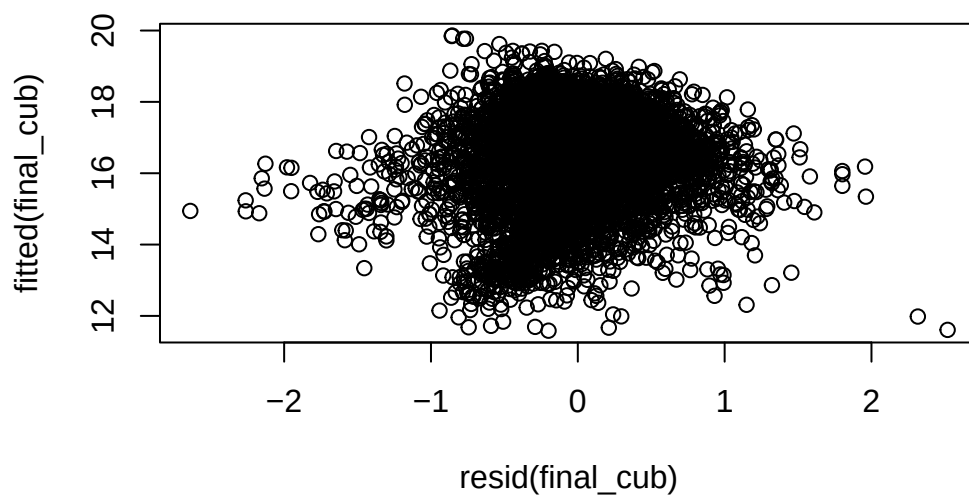
# Q-Q plot of residuals of the model

qqnorm(resid(final_cub))
qqline(resid(final_cub))

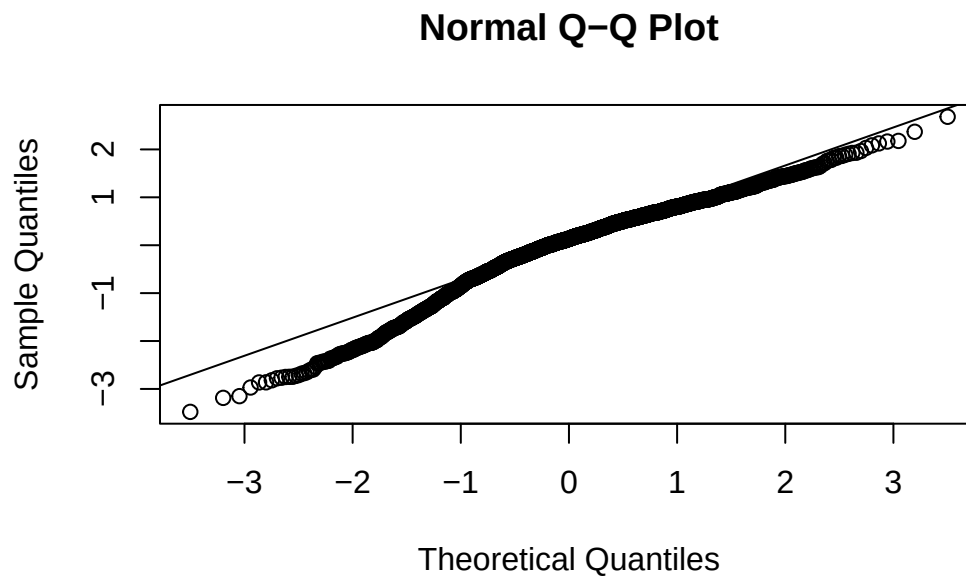
```



```
plot(resid(final_cub),fitted(final_cub))
```

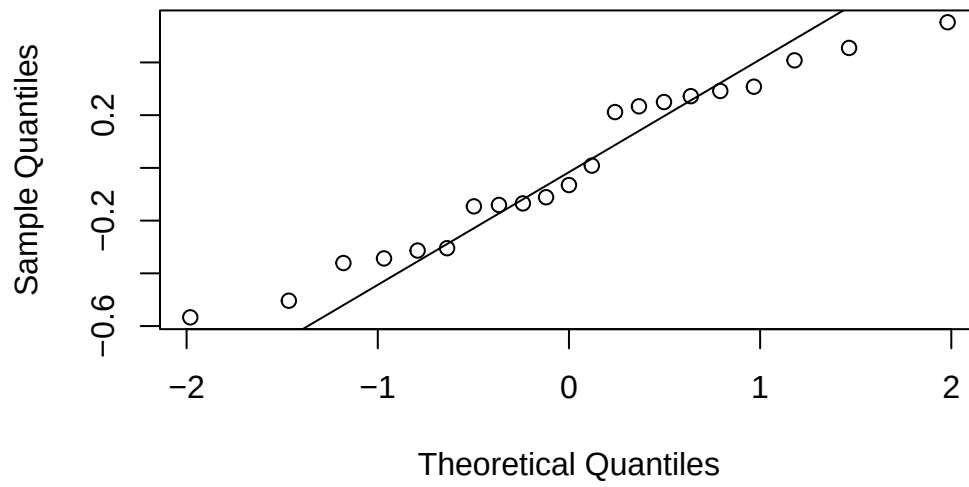


```
# slopes  
qqnorm(ranef(final_cub)$player_id[,1]) # group factor  
qqline(ranef(final_cub)$player_id[,1])
```



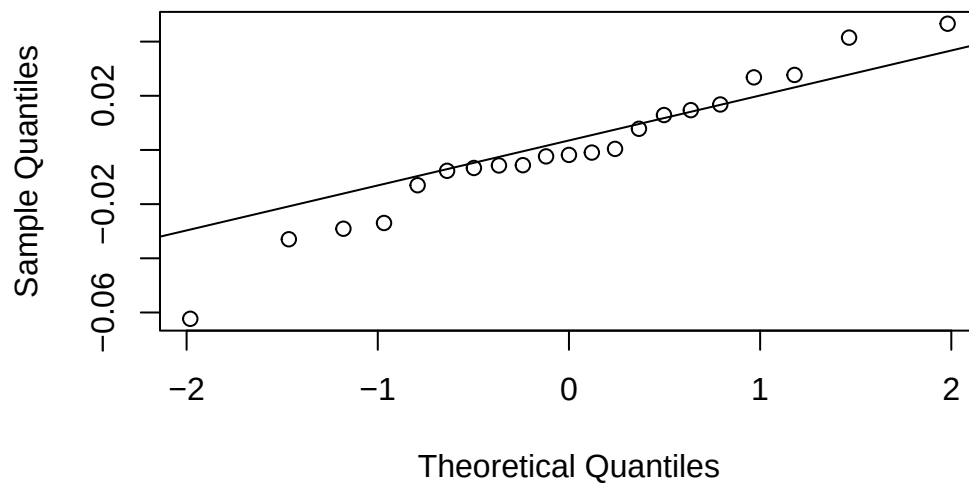
```
qqnorm(ranef(final_cub)$club_id[,1]) # group factor  
qqline(ranef(final_cub)$club_id[,1])
```

Normal Q-Q Plot



```
qqnorm(ranef(final_cub)$club_id[,2]) # group factor  
qqline(ranef(final_cub)$club_id[,2])
```

Normal Q-Q Plot



	M1	M2	M3
(Intercept)	15.981 ***	15.816 ***	15.547 ***
	[15.814, 16.148]	[15.678, 15.955]	[15.118, 15.977]
season__year__c	-0.029 **	0.011	0.010 ***
	[-0.047, -0.011]	[-0.001, 0.023]	[0.005, 0.016]
yellow__cards		0.002	-0.000
		[-0.005, 0.009]	[-0.007, 0.007]
red__cards		-0.012	-0.014
		[-0.060, 0.036]	[-0.062, 0.035]
goals		0.008 ***	0.007 **
		[0.004, 0.012]	[0.002, 0.011]
assists		0.012 ***	0.011 ***
		[0.006, 0.018]	[0.005, 0.017]
minutes__played		0.000 ***	0.000 ***
		[0.000, 0.000]	[0.000, 0.000]
age__c		-0.021 ***	-0.020 ***
		[-0.022, -0.020]	[-0.021, -0.020]
height__in__cm__c			0.012 ***
			[0.005, 0.019]
foreign__players			0.383 ***
			[0.194, 0.572]
uefa__coefficient__c			-0.001
			[-0.003, 0.002]
positionDefender			-0.327 ***
			[-0.440, -0.213]
positionGoalkeeper			-0.902 ***
			[-1.084, -0.720]
positionMidfield			-0.145 *
			[-0.257, -0.033]
RegionEurope & Central Asia			0.294
			[-0.101, 0.689]
RegionLatin America & Caribbean			0.550 **

		M4
(Intercept)		15.933 ***
		[15.482, 16.384]
season__year__c		0.005
		[-0.008, 0.018]
yellow__cards		0.002
		[-0.004, 0.009]
red__cards		-0.014
		[-0.062, 0.035]
goals		0.007 **
		[0.003, 0.011]
assists		0.014 ***
		[0.008, 0.019]
minutes__played__c		0.000 ***
		[0.000, 0.000]
age__c		-0.020 ***
		[-0.021, -0.020]
height__in__cm__c		0.013 ***
		[0.006, 0.020]
foreign__players		0.607 ***
		[0.399, 0.814]
uefa__coefficient__c		-0.001
		[-0.004, 0.001]
positionDefender		-0.322 ***
		[-0.437, -0.208]
positionGoalkeeper		-0.894 ***
		[-1.078, -0.710]
positionMidfield	28	-0.140 *
		[-0.254, -0.027]
RegionEurope & Central Asia		0.289
		[-0.111, 0.689]
RegionLatin America & Caribbean		0.542 **